

Philip S. Boonstra

Associate Professor

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Education

PhD, Biostatistics, University of Michigan (Ann Arbor)	2009 – 2012
MS, Biostatistics, University of Michigan (Ann Arbor)	2007 – 2009
BA, Mathematics and Political Science, Calvin University (formerly Calvin College) (Grand Rapids, MI)	2002 – 2006

Professional Affiliations

Associate Professor, with tenure (Dept. of Biostatistics, University of Michigan)	2020 – present
Research Associate Professor (Dept. of Biostatistics, University of Michigan)	2019 – 2020
Research Assistant Professor (Dept. of Biostatistics, University of Michigan)	2013 – 2019
Affiliated Faculty (Center for Cancer Biostatistics, University of Michigan)	2013 – present
Affiliate Member, Cancer Control and Population Sciences Program (University of Michigan Rogel Cancer Center)	2016 – present
Graduate Student Research Assistant (Dept. of Biostatistics, UM)	Intermittent, 2009 - 2012
Graduate Student Research Assistant (Kidney and Epidemiology Cost Center, UM) (Chronic Kidney Disease Surveillance Team, UM Group)	Summers 2008, 2009
Summer Research Assistant (Dept. of Mathematics and Statistics, Calvin College)	Summers 2004, 2006

Honors & Fellowships

Rackham Predoctoral Fellowship (University of Michigan)	2011 - 2012
Rackham Conference Travel Grant (UM)	2010, 2011, 2012
Rackham Regents' Fellowship (UM)	2009 - 2011
NIH Predoctoral Research in the Biosciences Training Grant (UM)	2007 - 2009
1st Place, Michigan Autumn Take Home (MATH) Challenge (Calvin College)	2005
Dean's List (GPA > 3.5/4.0) (Calvin)	2002 - 2006
Presidential Scholarship (Calvin)	2002 - 2006

Teaching

Traditional

Public Health 345* (University of Michigan; 2 credit hours)	Winters, 2025-2026
Undergraduate course teaching the principles and application of data visualization with applications to public health datasets	
* Newly developed course	
Biostatistics 699 (University of Michigan; 4 credit hours)	Winters, 2019-2026
Capstone course for 2nd-year MS students/1st year PhD students	
Biostatistics 619 (UM; 3 credit hours)	Winter 2018
Biostatistics 619 (2 credit hours)	Falls 2014-2016
Graduate course on the design, conduct, and analysis of clinical trials	
Biostatistics 644 (UM; 2 credit hours)	Fall 2015
Graduate course on statistical methods for cancer research	
Teaching Assistant / Grader (Calvin College)	2004 - 2005
for undergraduate courses in calculus, discrete mathematics, and linear algebra	

Non-traditional

Data Science for Health Research Specialization (Coursera; Massive Open Online Course Series)	Launched Sep 2023
Asynchronous three-course series for health data science learners (co-developed/co-taught with Professor Bhramar Mukherjee)	
https://www.coursera.org/specializations/data-science-for-health-research	
Biostatistics 591* (University of Michigan; 2 credit hours)	Springs, 2020-2023
Introduction to R programming course for online MPH students (hosted on Coursera)	
* Newly developed course	
Big Data Summer Institute (UM)	Summer 2017
Faculty mentor for 6 week undergraduate research program	

Publications

1. Conte ML, **Boonstra PS**, Caverly TJ, Fishstrom A, Flynn AJ, Taksler GB, and Friedman CP. "The potential for a benefit-estimating algorithm to improve recommendations for preventive services: comparing algorithm recommendations with those of primary care providers" In press at *Medical Decision Making*
2. Zaaqoq AM, Fox AD, **Boonstra PS**, Riddle C, Chen WT, Priest JR, Hyer S, Rycus P, Stead CM, and Tonna JE (2026) "The Extracorporeal Life Support Organization (ELSO) Registry Data Quality and Integrity Program" *ASAIO Journal* 72:111–116 <https://doi.org/10.1097/MAT.0000000000002660>
PMC 12829488
3. Wang C, Abdelrahman S, Geng X, Burgess A, Hu YS, Rauf MA, Kady N, Fu Y, Reilly TA, Maine IP, **Boonstra PS**, Sabitov K#, Murga-Zamalloa C and Wilcox RA (2026). "CDK9 is a dependency in GATA-3 driven and MCL-1 independent T-cell Lymphomas" *Blood Cancer Journal* 16: Article 9 <https://doi.org/10.1038/s41408-025-01427-1>
#Biostatistics graduate student mentee
PMC 12783670
4. Chase EC#, **Boonstra PS**, and Taylor JMG (2025) "A multiple imputation approach for cumulative incidence estimation" (2025) *The American Statistician* 79:291–301 <https://doi.org/10.1080/00031305.2025.2453674>
#Biostatistics graduate student mentee
PMC 12360614

5. **Boonstra PS** and Orozco del Pino P# (2025) "A comparison of some existing and novel methods for integrating historical models to improve estimation of coefficients in logistic regression" *Journal of the Royal Statistical Society: Series A* 188:46–67 <https://doi.org/10.1093/jrsssa/qnae093>
#Biostatistics graduate student mentee
PMC 11728056
6. Alexander PM, ..., Barbaro RP for the endorsing societies of **Extracorporeal Life Support Organization (ELSO)** ... (2024) "Definitions of adverse events associated with extracorporeal membrane oxygenation in children: results of an international Delphi process from the ECMO-CENTRAL ARC" *The Lancet Child & Adolescent Health* 8:773–780 [https://doi.org/10.1016/S2352-4642\(24\)00132-9](https://doi.org/10.1016/S2352-4642(24)00132-9)
PMC 12917940
7. **Boonstra PS**, Owen DR#, and Kang J (2024) "Shrinkage priors for isotonic probability vectors and binary data modeling" *Pharmaceutical Statistics* 23:540–556 <https://doi.org/10.1002/pst.2372>
#Graduate student mentee
PMC 11737611
8. Strohbehn GW, Stadler WM, **Boonstra PS**, and Ratain MJ (2024) "Optimizing the doses of cancer drugs after usual dose-finding" *Clinical Trials* 21:340–349 <https://doi.org/10.1177/17407745231213882>
9. Tonna JE, **Boonstra PS**, MacLaren G, Paden M, Brodie D, Anders M, Hoskote A, Ramanathan K, Hyslop R, Fanning JJ, Rycus P, Stead C, Barrett NA, Mueller T, Gómez RD, Kapoor PM, Fraser JF, Bartlett RH, Alexander PMA, and Barbaro RP, on behalf of the Extracorporeal Life Support Organization Member Centers (2024) "Extracorporeal Life Support Organization Registry International Report 2022: 100,000 survivors" *ASAIO* 70:131–143 <https://doi.org/10.1097/mat.0000000000002128>
PMC 10962646
10. Chase EC#, Taylor JMG, and **Boonstra PS** (2024) "Modeling data using horseshoe process regression" *Statistics in Medicine* 43: 817–832 <https://doi.org/10.1002/sim.9991>
#Biostatistics graduate student mentee
PMC 12930000
11. Foucar CE, Foley DH, Aldous J#, Burke PW, Pettit KR, Benitez LL, Perissinotti AJ, Marini BL, **Boonstra PS**, and Bixby D (2024) "Real-World Outcomes with Immunosuppressive Therapy for Aplastic Anemia in Patients Treated at the University of Michigan" *European Journal of Haematology* <https://doi.org/10.1111/ejh.14131> 112: 424–432 <https://doi.org/10.1111/ejh.14131>
#Biostatistics graduate student mentee
12. **Boonstra PS**, Tabarrok A, and Strohbehn GW (2023) "Targeted randomization dose optimization trials enable fractional dosing of scarce drugs" *PLOS One* 18: e0287511 <https://doi.org/10.1371/journal.pone.0287511>
PMC 10615276
13. Phillips T, Bond D, Takiar R, Kump K, Kandarpa M, **Boonstra PS**, Mayer T, Nachar V, Wilcox R, Carty S, Karimi Y, Nikolovska-Coleska Z, Kaminski R, Herrera A, Maddocks K, Popplewell K, and Danilov A (2023) "Adding venetoclax to lenalidomide and rituximab is safe and effective in patients with untreated mantle cell lymphoma" *Blood Advances* 7, 4518–4527 <https://doi.org/10.1182/bloodadvances.2023009992>
PMC 10425679
14. Delaney PG, Eisner ZJ, Thullah AH, Turay P, Kpawuru S, **Boonstra PS**, and Raghavendran K (2023) "Evaluating feasibility of a novel mobile emergency medical dispatch tool for lay first responder prehospital response coordination in Sierra Leone: A simulation-based study" *Injury* 54, 5–14 <https://doi.org/10.1016/j.injury.2022.10.010>
15. Flynn AJ, Beck A, Boisvert P, **Boonstra PS**, Caverly T, Gittlen N, Meng G, Rath B, Taksler G, and Friedman CP (2023) "CBK model composition using paired web services and executable functions: A demonstration for individualizing preventive services" *Learning Health Systems* 7, e10325 <https://doi.org/10.1002/lrh2.10325>
PMC 10091204
16. Hinton T, Karnak D, Tang M, Jiang R, Luo Y, **Boonstra PS**, Sun Y, Nancarrow DJ, Sandford E, Ray P, Maurino C, Matuszak M, Green MD, Schipper MJ, Yanik GA, Tewari M, El Naqa I, Schonewolf CA, Ten Haken R, Jolly S,

- Lawrence TS, and Ray D (2022) "Improved prediction of radiation pneumonitis by combining biological and radiobiological parameters using a data-driven Bayesian network analysis" *Translational Oncology* 21, 101428 <https://doi.org/10.1016/j.tranon.2022.101428>
PMC 9046881
17. Kohne JG, MacLaren G, Cagino L, **Boonstra PS**, Brodie D, and Barbaro RP (2022) "Tracheostomy practices and outcomes in patients with COVID-19 supported by ECMO: An analysis of the ELSO registry" *Critical Care Medicine* 50, 1360–1370 <https://doi.org/10.1097/ccm.0000000000005579>
PMC 9380152
 18. Li P, Taylor JMGT, **Boonstra PS**, Lawrence TS, and Schipper MJ (2022) "Utility based approach in individualized optimal dose selection using machine learning methods" *Statistics in Medicine* 41, 2957–2977 <https://doi.org/10.1002/sim.9396>
PMC 9233043
 19. Barbaro RP*, MacLaren G*, **Boonstra PS**, Combes A, Agerstrand C, Annich G, Diaz R, Fan E, Hryniewicz K, Lorusso R, Paden ML, Stead CM, Swol J, Iwashyna TJ, Slutsky AS, Brodie D, for the Extracorporeal Life Support Organization (2021) "Extracorporeal Membrane Oxygenation for COVID-19: Evolving outcomes from an international registry" *The Lancet* 398, 1230–1238 [https://doi.org/10.1016/S0140-6736\(21\)01960-7](https://doi.org/10.1016/S0140-6736(21)01960-7)
PMC 8480964
 20. West BT, Little RJA, Andridge RR, **Boonstra PS**, Ware EB, Pandit A, and Alvarado-Leiton F (2021) "Assessing selection bias in regression coefficients estimated from nonprobability samples with applications to genetics and demographic surveys" *Annals of Applied Statistics* 15, 1556–1581 <https://doi.org/10.1214/21-AOAS1453>
PMC 8887878
 21. **Boonstra PS**, Little RJA, West BT, Andridge RR, and Alvarado-Leiton F (2021) "A simulation study of diagnostics for bias in non-probability samples" *Journal of Official Statistics* 37, 751–769 <https://doi.org/10.2478/jos-2021-0033>
PMC 8460089
 22. Delaney PG, Eisner ZJ, Thullah AH, Muller B, Gan J, Timbo A, Kamara S, Sandy K, **Boonstra PS**, Scott JW, Raghavendran K (2021) "Evaluating a novel Prehospital Emergency Trauma Care Assessment Tool (PETCAT) for low- and middle-income countries: Lay first responders expand prehospital care in Sierra Leone" *World Journal of Surgery* 45, 2370–2377 <https://doi.org/10.1007/s00268-021-06140-1>
 23. **Boonstra PS**, Braun TM, and Chase EC# (2021) "A modular framework for seamless oncology trials" *Clinical Trials* 18, 303–313 <https://doi.org/10.1177/1740774520981939>
#Biostatistics graduate student mentee
PMC 8382494
 24. Salami SS*, Tosoian JJ*, Srinivas N, Jones, Jr TA, Brockman S, Elkhoury FF, Bazzi S, Plouffe KR, Sumida L, Siddiqui J, Liu CJ, Kunju LP, Morgan TM, Natarajan S, **Boonstra PS**, Sumida L, Tomlins SA, Udager AM, Sisk, Jr AE, Marks LS, and Palapattu GS (2021) "Serial molecular profiling of low-grade prostate cancer to assess tumor upgrading: A longitudinal cohort study" *European Urology* 79, 456–465 <https://doi.org/10.1016/j.eururo.2020.06.041>
*Equal contribution authors
PMC 7779657
 25. Owen DR#, Sun Y, **Boonstra PS**, McFarlane M, Viglianti BL, Balter JM, El Naqa I, Schipper MJ, Schonewolf CA, Ten Haken RK, Kong FM, Jolly S, and Matuszak MM (2021) "Investigating the SPECT dose-function metrics associated with radiation-induced lung toxicity risk in non-small cell lung cancer patients undergoing radiation therapy" *Advances in Radiation Oncology* 6, 100666 <https://doi.org/10.1016/j.adro.2021.100666>
#Graduate student mentee
PMC8010578
 26. **Boonstra PS**, Ahmed A, Merrill SA, and Wilcox RA (2021) "Ruxolitinib in adult patients with secondary hemophagocytic lymphohistiocytosis" *American Journal of Hematology* 96: E103-E105 <https://doi.org/10.1002/ajh.26091>
PMC 8968463

27. Little RJA, West BT, **Boonstra PS**, and Hu J (2020) "Measures of the degree of departure from ignorable sample selection" *Journal of Survey Statistics and Methodology* 8, 932–964 <https://doi.org/10.1093/jssam/smz023>
PMC 7750890
28. Barbaro RP*, MacLaren G*, **Boonstra PS**, Iwashyna TJ, Slutsky AS, Fan E, Paden ML, Ogino MT, Bartlett RH, Tonna J, Hyslop R, Fanning JJ, Rycus PT, Hyer S, Anders M, Agerstrand C, Hryniewicz K, Diaz R, Lorusso R, Combes A, Brodie D, for the Extracorporeal Life Support Organization (2020) "Extracorporeal membrane oxygenation support in COVID-19: The registry report from the Extracorporeal Life Support Organization" *The Lancet* 396, 1071–1078 [https://doi.org/10.1016/S0140-6736\(20\)32008-0](https://doi.org/10.1016/S0140-6736(20)32008-0)
*Equal contribution authors
PMC 7518880
29. **Boonstra PS** and Krauss JC (2020) "Inferring a consensus problem list using penalized multistage models for ordered data" *Annals of Applied Statistics* 14, 1557–1580 <https://doi.org/10.1214/20-AOAS1361>
PMC 8345315
30. Baker LH, **Boonstra PS**, Reinke D, Peregrine Antalis E, Zebrack B, and Weinberg RL "Burden of chronic diseases among sarcoma survivors treated with anthracycline chemotherapy: Results from an observational study" (2020) *Journal of Cancer Metastasis and Treatment* 6, 24 <https://doi.org/10.20517/2394-4722.2020.36>
PMC 8513741
31. Bankhead A, McMaster T, **Boonstra PS**, and Palmboos PL (2020) "Isoform-specific TP63 expression is linked with distinct clinical outcomes in bladder cancer" *EBioMedicine* 51, 102561 <https://dx.doi.org/10.1016/j.ebiom.2019.11.022>
PMC 6953644
32. Hanks JE, Kovatch KJ, Ali SA, Roberts E#, Durham AB, Smith JD, Bradford CR, Malloy KM, **Boonstra PS**, and McLean SA (2020) "Long-term outcomes, prognostic value, and accuracy of sentinel lymph node biopsy in head and neck melanoma" *Otolaryngology-Head and Neck Surgery* 162, 520–529 <https://doi.org/10.1177/0194599819899934>
#Biostatistics graduate student mentee
PMC 8012010
33. **Boonstra PS** and Barbaro RP (2020) "Incorporating historical models with adaptive Bayesian updates" *Biostatistics* 21, e47–e64 <https://doi.org/10.1093/biostatistics/kxy053>
PMC 7868052
34. Ahmed A, Merrill SA, Alsawah F, Bockenstedt P, Campagnaro E, Devata S, Gitlin SD, Kaminski M, Cusick A, Phillips T, Sood S, Talpaz M, Quiery A, **Boonstra PS**, and Wilcox RA (2019) "Ruxolitinib as a targeted therapy in adult patients with secondary HLH" *Lancet Haematology* 6, e630–e637 [https://doi.org/10.1016/S2352-3026\(19\)30156-5](https://doi.org/10.1016/S2352-3026(19)30156-5)
PMC 8054981
35. Andridge R, West BT, Little RJA, **Boonstra PS**, and Alvarado-Leiton F (2019) "Indices of non-ignorable selection bias for proportions estimated from non-probability samples" *Journal of the Royal Statistical Society: Series C* 68, 1465–1483 <https://doi.org/10.1111/rssc.12371>
PMC 7724611
36. Chase EC# and **Boonstra PS** (2019) "Accounting for established predictors with the multi-step elastic net" *Statistics in Medicine* 38, 4534–4544
#Biostatistics graduate student mentee
PMC 6736688
37. Geer M*, Roberts E*#, Shango M, Till B, Smith S, Abbas H, Hill B, Kaplan J, Barr PM, Caimi PF, Stephens DM, Lin E, Herrera AF, Rosenbaum E, Amengual K, **Boonstra PS**, Devata S, Wilcox RA, Kaminski M, and Phillips T (2019). "Multicenter retrospective study of intravascular DLBCL treated at academic institutions within the United States" *British Journal of Haematology* 186, 255–262 <https://doi.org/10.1111/bjh.15923>
*Equal contribution authors
#Biostatistics graduate student mentee
38. Menghrajani K*, **Boonstra PS***, Perkins C, Gowin KL, Mesa RA, Gotlib JR, Wang L, Singer JW, and Talpaz, M (2019). "Predictive models for splenic response to JAK2 inhibitor therapy in patients with myelofibrosis"

Leukemia & Lymphoma 60, 1036–1042 <https://doi.org/10.1080/10428194.2018.1509315>

*Equal contribution authors

PMC 6426689

39. **Boonstra PS**, Barbaro RP, and Sen A (2019). “Default priors for the intercept in logistic regressions” *Computational Statistics & Data Analysis* 133, 245–256 <https://doi.org/10.1016/j.csda.2018.10.014>
PMC 6748335
40. Barbaro RP, **Boonstra PS**, Kuo KW, Selewski DT, Bailly DK, Stone CL, Chow J, Annich GM, Moler FW, and Paden ML (2019). “Evaluating mortality risk adjustment among children receiving extracorporeal support for respiratory failure” *ASAIO Journal* 65, 277–284 <https://doi.org/10.1097/mat.0000000000000813>
41. Owen DR, **Boonstra PS**, Viglianti BL, Balter JM, Schipper MJ, Jackson W, El Naqa I, Jolly S, Ten Haken RK, and Matuszak MM (2018). “Modeling patient-specific dose-function response using SPECT/CT to personalize prediction of radiation-induced lung toxicity” *International Journal of Radiation Oncology · Biology · Physics* 102, 1265–1275 <https://doi.org/10.1016/j.ijrobp.2018.05.049>
PMC 6202237
42. Ma VT, **Boonstra PS**, Menghrajani K, Perkins C, Gowin KL, Mesa RA, Gotlib JR, and Talpaz, M (2018). “Treatment with JAK inhibitors in myelofibrosis patients nullifies the prognostic impact of unfavorable cytogenetics” *Clinical Lymphoma, Myeloma, and Leukemia* 18, e201–e210.
PMC 5927833
43. Hawkins PG*, **Boonstra PS***, Hobson ST, Hayman JA, Ten Haken RK, Matuszak MM, Stanton P, Kalemkerian GP, Lawrence TS, Schipper MJ, and Jolly S (2018). “Prediction of radiation esophagitis in non-small cell lung cancer using clinical factors, dosimetric parameters, and pretreatment cytokine levels.” *Translational Oncology* 11, 102–108.
*Equal contribution authors
PMC 6002355
44. **Boonstra PS***, Polk A*, Brown B, Hristov AC, Bailey NG, Lu Y, Kaminski MS, Phillips T, Devata S, Mayer T, and Wilcox RA (2017). “A single center phase II study of ixazomib in patients with relapsed or refractory cutaneous or peripheral T-cell lymphomas” *American Journal of Hematology* 92, 1287–1294.
*Equal contribution authors
PMC 6116510
45. Grivas PD, Devata S, Khoriaty R, **Boonstra PS**, Ruch J, McDonnell K, Hernandez LA, Wilfong J, Smerage J, Ison MG, Eisenberg JNS, Silveira M, Cooney KA, and Worden FP (2017). “Low-cost intervention to increase influenza vaccination rates at a comprehensive cancer center” *Journal of Cancer Education* 32, 871–877.
46. von Salomé J, **Boonstra PS**, Karimi M, Aravidis C, Silander G, Stenmark-Askmal M, Nilbert M, Lindblom A, and Lagerstedt-Robinson K (2017). “Genetic anticipation in Swedish Lynch syndrome families” *PLOS Genetics* 13, e1007012.
PMC 5681299
47. Hawkins PG*, **Boonstra PS***, Ten Haken RK, Matuszak MM, Kong FM, Hearn JM, Lawrence TS, Schipper MJ, and Jolly S (2017). “Radiation-induced lung toxicity in non-small-cell lung cancer: Understanding the interactions of clinical factors and cytokines with the dose-toxicity relationship” *Radiotherapy and Oncology* 125, 66–72.
PMC 5645039
*Equal contribution authors
48. Barbaro RP, **Boonstra PS**, Moler FW, Paden ML, Davis MM, Prosser LA (2017). “Hospital-level variation in inpatient costs among children receiving extracorporeal membrane oxygenation” *Perfusion* 32, 538–546.
49. DaSilva AF, Nascimento TD, Jassar H, Heffernan J, Toback RL, Lucas S, DosSantos MF, Bellile EL, **Boonstra PS**, Taylor JMG, Casey KL, Koeppe RA, Smith YR, and Zubieta JK (2017) “Dopamine D2/D3 imbalance during migraine attack and allodynia in vivo” *Neurology* 88, 1634–1641.
PMC 5405765
50. Moyer JS, Flynn S, **Boonstra PS**, Kraft C, Chinn S, Baker SR, Schwartz JL, Bichakjian, CK, Fullen D, Durham AB, Lowe L, Johnson TM (2017). “Efficacy of staged excision with permanent section margin control for cutaneous head and neck melanoma” *JAMA Dermatology* 153, 282–288.

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52. **Boonstra PS**, Braun TM, Taylor JMG, Kidwell KM, Bellile EL, Daignault S, Zhao L, Griffith KA, Lawrence TS, Kalemkerian GP, and Schipper MJ (2017). "Statistical controversies in cancer research: Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts" *Annals of Oncology* 28, 1427–1435.
PMC 5834117
53. Polk A, Lu Y, Wang T, Seymour E, Bailey NG, Singer JW, **Boonstra PS**, Lim MS, Malek S, and Wilcox RA (2016). "Colony-stimulating Factor-1 Receptor is required for nurse-like cell survival in chronic lymphocytic leukemia" *Clinical Cancer Research* 22, 6118–6128.
PMC 5161678
54. **Boonstra PS**, Taylor JMG and Mukherjee B (2016). "Increasing efficiency for estimating treatment-biomarker interactions with historical data" *Statistical Methods in Medical Research* 25, 2959–2971.
PMC 5450810
55. Barbaro RP, **Boonstra PS**, Paden ML, Folafoluwa OO, Roberts LA, Annich GM, Bartlett RH, Moler FW, Davis MM (2016). "Development and validation of pediatric risk estimate score for children using extracorporeal respiratory support (Ped-RESCUERS)" *Intensive Care Medicine* 42, 879–888.
56. Krauss JC, **Boonstra PS**, Vantsevich A, Friedman CP (2016) "Is the problem list in the eye of the beholder?: An exploration of consistency across physicians" *Journal of the American Medical Informatics Association* 23, 859–865.
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57. **Boonstra PS**, Taylor JMG, Smolska-Ciszewska B, Behrendt K, Dworzecki T, Gawkowska-Suwinska M, Bialas B, and Suwinski R (2016). " α/β ratio for prostate cancer derived from external beam radiotherapy and brachytherapy boost" *The British Journal of Radiology* 89, 20150957.
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58. **Boonstra PS**, Mukherjee B, Gruber SB, Ahn J, Schmit SL, and Chatterjee N (2016). "Tests for gene-environment interactions and joint effects with exposure misclassification" *American Journal of Epidemiology* 183, 237–247.
PMC 4724093
59. **Boonstra PS**, Taylor JMG and Mukherjee B (2015). "Adaptive shrinkage via the hyperpenalized EM algorithm" *Statistics in Biosciences* 7, 417–431.
PMC 4728141
60. **Boonstra PS**, Mukherjee B and Taylor JMG (2015). "A small-sample choice of the tuning parameter in ridge regression" *Statistica Sinica* 25, 1185–1206.
PMC 4790465
61. **Boonstra PS**, Shen JC, Taylor JMG, Braun TM, Griffith KA, Daignault S, Kalemkerian GP, Lawrence TS, and Schipper MJ (2015). "A statistical evaluation of dose expansion cohorts in phase I clinical trials" *Journal of the National Cancer Institute* 107, dju429.
PMC 4565529
62. Stenzel SL, Ahn J, **Boonstra PS**, Gruber SB, and Mukherjee B (2015). "The impact of exposure-biased sampling designs on detection of gene-environment interactions in case-control studies with potential exposure misclassification" *European Journal of Epidemiology* 30, 413–423.
PMC 4256150
63. **Boonstra PS***, Bondarenko I*, Park SK, Vokonas PS and Mukherjee B (2014). "Propensity score-based diagnostics for categorical response regression models" *Statistics in Medicine* 33, 455–469.
PMC 3911784
*Equal contribution authors
64. **Boonstra PS**, Mukherjee B and Taylor JMG (2013). "Bayesian shrinkage methods for partially observed data with many predictors" *The Annals of Applied Statistics*, 7, 2272–2292.
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65. **Boonstra PS**, Taylor JMG and Mukherjee B (2013). "Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches" *Biostatistics* 14, 259–272.
PMC 3590922
66. **Boonstra PS**, Mukherjee B, Taylor JMG, Nilbert M, Moreno V and Gruber SB (2011). "Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in hereditary non-polyposis colorectal cancer (Lynch syndrome)" *Biometrics* 67, 1627–1637.
PMC 3176998
67. **Boonstra PS**, Gruber SB, Raymond VM, Huang SC, Timshel S, Nilbert M and Mukherjee B (2010). "A review of statistical methods for testing genetic anticipation: Looking for an answer in Lynch syndrome" *Genetic Epidemiology* 34, 756–768.
PMC 3894615
68. Saran R, Hedgeman E, Plantinga L, Burrows NR, Gillespie BW, Young EW, Coresh J, Pavkov M, Williams D and Powe NR, for the **CKD Surveillance Team** (2010). "Establishing a National Chronic Kidney Disease Surveillance System for the United States" *Clinical Journal of the American Society of Nephrology* 5, 152–161.

Submitted / under revision

- Kohne JG, **Boonstra PS**, Dempsey W, Alexander PMA, Friedman ML, Horvat CM, MacLaren G, Raman L, Rintoul N, Vogel AM, and Barbaro RP. "Impact of Changing Cannulation Strategy on Neurologic Injury in Infants with Respiratory Failure"
- Rali AS, Sarma D, Li H, Urner M, **Boonstra PS**, Alexander PMA, Barbaro RP, Alviar C, Jentzer J, Grandin WE, Bacchetta MD, Lima B, Shah Z, Rycus P, Tonna JE "Prognostic implications of complications during VA ECMO for cardiogenic shock"
- Gutierrez MG, Wang C, Inamdar K, Stone MB, Ehyae V, Venkataraman G, **Boonstra PS**, Murga-Zamalloa C, and Wilcox RA "GATA-3 identifies a distinct subset of T-cell prolymphocytic leukemia (T-PLL) and is associated with CCR4 expression"
- Geng X, Abdelrahman S, Wang C, Kady N, Maine IP, Yang Y, Wang S, Robida A, **Boonstra PS**, Lee J, Reneau J, Murga-Zamalloa C, and Wilcox RA (submitted) "BCL-xL is a therapeutic vulnerability in T-cell receptor and GATA-3 driven non-ALCL T-cell lymphomas"
- Gutierrez MG, Hemenway N, **Boonstra PS**, and Wilcox RA "A review of early-phase trials for the treatment of PTCL for informing future designs"

Conference Abstracts / Posters

1. Gutierrez M, Zamalloa CM, Inamdar K, Stone M, Ehyae V, **Boonstra PS**, and Wilcox R (2025) "GATA-3 identifies a distinct subset of T-cell prolymphocytic leukemia (T-PLL) and is associated with CCR4 expression" *Blood* 146:1877
2. Phillips TJ, Weiss J, Bond D, Xiao A, Lopedote P, Karimi Y, ..., **Boonstra PS**, Mayer T, and Danilov A (2025) "Long-term follow up of the VALOR (venetoclax, lenalidomide, rituximab) in patients with untreated mantle cell lymphoma" *Hematological Oncology* 43:e135.70093
3. Ye JC, **Boonstra PS**, Campagnaro EL, Pianko M, Kin A, Anderson Jr LD, et al. and Zonder JA (2024) "Phase 2 study using minimal residual disease-driven adaptive strategy in newly diagnosed multiple myeloma with upfront daratumumab-based therapy—first report from the Maestro trial" *Blood* 144:3379
4. Takiar R, Ng WL, Shah E, **Boonstra PS**, Karimi YH, Carty SA, et al. and Phillips TJ (2022) "A multi-center study: outcomes among relapsed/refractory diffuse large B cell lymphoma patients with CD20 loss" *Blood* 140: 9510–9511.
5. Takiar R, **Boonstra PS**, Nachar V, and Phillips T (2022) "ABCL-338 impact of DA-R-EPOCH for high-grade B-cell lymphoma patients at the University of Michigan" *Clinical Lymphoma Myeloma and Leukemia* 22:S373–S374

6. Takiar R, **Boonstra PS**, Karimi Y, Carty S, Wilcox RA, and Phillips TJ (2022) "A real-world experience: outcomes among relapsed/refractory diffuse large B cell lymphoma patients with CD20 loss" *J Clin Oncol* 40:e19570
7. Ye JC, **Boonstra PS**, Boyer DF, Anderson LD, Lipe BC, Kin A, et al. and Talpaz M (2019) "A phase 2 study with minimal residual disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma with upfront daratumumab-based therapy" *Blood* 134:3191

Student Mentoring / Advising

PhD Advisor/Co-Advisor

PhD Committee Co-chair: Wei-Hsiang (Sean) Sun Department of Biostatistics, UM	2028 (anticipated)
PhD Committee Co-chair: Rachel Tucker Gonzalez Department of Biostatistics, UM	2027 (anticipated)
PhD Committee Co-chair: Tian Xie Department of Biostatistics, UM	2025 (withdrew)
PhD Committee Co-chair: Elizabeth Chase Department of Biostatistics, UM	2023

PhD Committee Member

PhD Committee Member: Sergio Martinez Program in Survey and Data Science, UM	2027 (anticipated)
PhD Committee Member: Sarah Ferlito Department of Biostatistics, UM	2026 (anticipated)
PhD Committee Member: Nathaniel Osher Department of Biostatistics, UM	2025
PhD Committee Member (Cognate): Ramya Naraharisetti Department of Epidemiology, UM	2023
PhD Committee Member: Pedro Orozco del Pino Department of Biostatistics, UM	2022
PhD Committee Member (Cognate): Ali Rafei Survey Methodology Program, UM	2021
PhD Committee Member (Cognate): Daniel Rocky Owen Department of Nuclear Engineering & Radiological Sciences, UM	2021
PhD Committee Member: Pin Li Department of Biostatistics, UM	2020
PhD Committee Member: Zhichao Sun Department of Biostatistics, UM	2016

Graduate Student Research Assistant Advisor

Wei-Hsiang (Sean) Sun	2023-2026
Qiyuan Shi	2022-2023
Elizabeth Chase	2021-2022
Selena Scott	2021-2022
Pedro Orozco del Pino	2020-2022

T32 Project Mentor

John Lee ('29 [anticipated]); Nathan Hemenway ('28 [anticipated]); Rachel Tucker Gonzalez ('27 [anticipated]); Elizabeth Chase ('23); Emily Roberts ('21);

MS Academic Advisor

Ishan Ghosh ('27 [anticipated]); Jintong Hong ('26); Kirill Sabitov ('25); Xinwei Wang ('25); Su Xu ('25); Chongwei Shi ('24); Ebo Essilfie-Amoah (withdrew '24); Yahui Zhang ('23); Weizhou Qian ('23); Haisheng Xu ('23); Grace Tiernon ('23); Sahiti Kolli ('21); Zhongzhe Ouyang ('21); Chenyi Yu (withdrew '21); Jane Li ('20); Katherine Li ('20); Chun-Hui Lin ('20); Evan Wiewiora ('20); Michelle Earley ('15); Marco Benedetti ('15)

Software

isotonicBayes: R package implementing the isotonic probability vectors in Boonstra, Owen, and Kang (2023). Available at <https://github.com/psboonstra/isotonicBayes>

DoseDeescalation: R package implementing and simulating the dose-deescalation designs as proposed in Boonstra, Tabarrok, and Strohbehn (2023). Available at <https://github.com/psboonstra/DoseDeescalation>

healthds: R package to accompany the 'Data Science for Health Research' Coursera Specialization. Available at <https://github.com/umich-biostatistics/healthds>

seamlessim: R package implementing a simulator for modular seamless oncology trials as proposed in Boonstra, Braun, and Chase (2021). Available at <https://github.com/elizabethchase/seamlessim>. Co-authored with Elizabeth C. Chase

adaptBayes: R package implementing the adaptive Bayesian priors proposed in Boonstra and Barbaro (2020). Available at <https://github.com/umich-biostatistics/adaptBayes>. Co-authored with Mike Kleinsasser

Clinical Trials

Completed / Terminated

1. "Open-label, single-center phase II study of MLN9708 (Ixazomib) in patients with relapsed/refractory cutaneous and peripheral T-cell lymphomas" PI: R Wilcox; [clinicaltrials.gov: NCT02158975](https://clinicaltrials.gov/ct2/show/study/NCT02158975)
2. "Phase Ib, open label, single center study of pacritinib in relapsed/refractory lymphoproliferative disorders" PI: R Wilcox; [clinicaltrials.gov: NCT03601819](https://clinicaltrials.gov/ct2/show/study/NCT03601819)
3. "A pilot study of ruxolitinib in secondary hemophagocytic syndrome" PI: R Wilcox; [clinicaltrials.gov: NCT02400463](https://clinicaltrials.gov/ct2/show/study/NCT02400463)
4. "Phase I dose escalation study to determine the maximum tolerated dose of the combination of ruxolitinib and bortezomib in patients with relapsed or refractory lymphoma" PI: T Phillips; [clinicaltrials.gov: NCT02613598](https://clinicaltrials.gov/ct2/show/study/NCT02613598)
5. "A phase I/II study of ruxolitinib in combination with nilotinib in patients with chronic phase CML who have achieved a major molecular remission but not a complete molecular remission on a tyrosine kinase inhibitor alone" PI: P Burke; [clinicaltrials.gov: NCT02973711](https://clinicaltrials.gov/ct2/show/study/NCT02973711)
6. "Phase II study of nivolumab and the antagonistic CSF-1R monoclonal antibody cabiralizumab (BMS-986227) in patients with relapsed/refractory peripheral T cell lymphoma" PI: R Wilcox; [clinicaltrials.gov: NCT03927105](https://clinicaltrials.gov/ct2/show/study/NCT03927105)
7. "Phase I/II study of ixazomib and romidepsin in relapsed/refractory peripheral T-cell lymphoma (PTCL)" PI: S Devata; [clinicaltrials.gov: NCT03547700](https://clinicaltrials.gov/ct2/show/study/NCT03547700)
8. "Phase II multi-center single arm study evaluating the effect of adding bortezomib to ibrutinib in ibrutinib relapsed mantle cell lymphoma" PI: T Phillips; [clinicaltrials.gov: NCT03617484](https://clinicaltrials.gov/ct2/show/study/NCT03617484)

9. "Phase IB dose de-escalation study of the PI3k alpha/delta inhibitor, copanlisib given in combination with the immunotherapeutic agents, nivolumab and rituximab in patients with relapsed/refractory indolent lymphoma LYM17-145" PI: T Phillips; clinicaltrials.gov: NCT04431635
10. "A phase II study using a BFM regimen plus tyrosine kinase inhibitor in adult Philadelphia chromosome-positive acute lymphoblastic leukemia" PI: P Burke; clinicaltrials.gov: NCT04845035

Active / Approved

11. "A Pilot Study of the Addition of Venetoclax to Standard Remission Induction Chemotherapy Fludarabine or Cladribine, Cytarabine, and Granulocyte Colony Stimulating Factor (G-CSF) (FLAG or CLAG) for Frontline Therapy of Secondary AML" PI: D Bixby; clinicaltrials.gov: NCT05780879
12. "A phase II pilot study of metformin therapy in patients with relapsed chronic lymphocytic leukemia and untreated CLL patients with genomic deletion 11q" PI: S Malek; clinicaltrials.gov: NCT01750567
13. "Phase I multi-site study evaluating the MTD, safety and efficacy of the combination venetoclax, lenalidomide and rituximab in patients with previously untreated mantle cell lymphoma" PI: Sano; clinicaltrials.gov: NCT03523975
14. "Phase 2 study with minimal residual disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma with upfront daratumumab-based therapy" PI: C Ye; clinicaltrials.gov: NCT04140162
15. "Phase II, open label, multicenter study of pacritinib in relapsed/refractory T-cell lymphoproliferative neoplasms" PI: R Wilcox; clinicaltrials.gov: NCT04858256
16. "A phase II study of pembrolizumab and mogamulizumab in advanced-stage, relapsed/refractory cutaneous T-cell lymphomas" PI: R Wilcox; clinicaltrials.gov: NCT05956041

Funding

Effort is annualized calendar months

Submitted

PCORI (PI Barbaro); \$20,653,539 <i>Comparing Accessory-Rich Versus Streamlined Extracorporeal Membrane Oxygenation Circuits to Reduce Adverse Events in Children</i> https://www.pcori.org/funding-opportunities/announcement/phased-large-awards-comparative-effectiveness-research-pcori-funding-announcement-cycle-3-2025 Role: Co-I (2.8 calendar months)	01/01/27 – 06/30/33
NIH/NCI (PI Wilcox); \$2,529,141 <i>R01: Novel Multi-targeted Strategies Targeting T-cell Lymphomas and their Microenvironment</i> https://grants.nih.gov/grants/guide/pa-files/PA-25-301.html Role: Co-I (0.60 calendar months)	12/01/25 – 11/30/30
NIH/NCI (PI Wilcox); \$2,649,646 <i>R01: Targeting the GATA-3 Regulon in T-cell lymphomas</i> https://grants.nih.gov/grants/guide/pa-files/pa-20-185.html Role: Co-I (0.60 calendar months)	07/01/25 – 06/30/30

Current

Extracorporeal Life Support Organization (PI Boonstra); \$691,880 <i>ELSO Registry Contract</i> Role: PI (4.2 calendar months)	07/01/22 – 06/30/26
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2-P30CA046592, NIH/NCI (PI Fearon); \$25,866,261 <i>Comprehensive Cancer Center Core Grant II Biostatistics Core</i> Role: Co-I; Member, Biostatistics Core (0.86 calendar months)	06/01/12 – 05/31/28
5-R01CA265929 , NIH/NCI (PI Wilcox); \$2,512,985 <i>Pacritinib in rel/refr T-cell lymphomas</i> Role: Co-I (0.60 calendar months)	9/01/21 – 8/31/27
1R01CA278976, NIH/NCI (PI Wilcox); \$3,119,617 <i>Notch ligands as oncogenic drivers and therapeutic targets in T-cell lymphomas</i> Role: Co-I (0.30 calendar months)	1/01/24 – 12/31/28
CTSU-2022.116 Wilcox UM MS IIT with HCRN (Merck); \$375,645 <i>A phase II study of pembrolizumab and mogamulizumab in advanced-stage, relapsed/refractory cutaneous T-cell lymphomas</i> Role: Co-I (0.06 calendar months)	10/16/23 – 10/11/29

Completed

1-R01CA236722, NIH/NCI (PI Wilcox); \$1,250,000 <i>Notch and GATA-3 as novel therapeutic targets in T-cell lymphomas</i> Role: Co-I	12/01/19 – 11/30/24
Clinical Trial (PI Phillips); \$307,438 Hoosier Cancer Research Network/Bristol-Myers Squibb Company <i>Phase IB dose de-escalation study of the PI3k alpha/delta inhibitor, Copanlisib given in combination with the immunotherapeutic agents, Nivolumab and Rituximab in patients with relapsed/refractory indolent lymphoma LYM17-145</i> Role: Biostatistician	03/03/20 – 3/28/24
Clinical Trial (PI Ye); \$782,241 Janssen Scientific <i>Phase 2 study with Minimal Residual Disease (MRD) driven adaptive strategy in treatment for newly diagnosed multiple myeloma (MM) with upfront daratumumab-based therapy</i> Role: Biostatistician	10/22/19 – 8/11/23
5-R01CA129102, NIH/NCI (PIs Taylor/Ghosh); \$1,568,291 <i>Statistical Methods for Cancer Biomarkers</i> Role: Co-I	01/01/09 – 06/30/22
2-T32CA083654, NIH/NCI (PI Taylor); \$192,805 (FY19) <i>Biostatistics Training In Cancer Research</i> Role: Program Associate Director	06/01/12 – 03/31/23
Clinical Trial (PI Wilcox); \$567,436 Hoosier Cancer Research Network/Bristol-Myers Squibb Foundation <i>Phase II study of nivolumab and the antagonistic CSF-1R monoclonal antibody cabiralizumab (BMS-986227) in patients with relapsed/refractory peripheral T cell lymphoma</i> Role: Biostatistician	04/16/19 – 03/31/22
Clinical Trial (PI Devata); \$294,967 Hoosier Cancer Research Network/Millennium Pharmaceuticals, Inc. <i>Phase I/II study of ixazomib and romidepsin in relapsed/refractory peripheral T-cell lymphoma (PTCL)</i> Role: Biostatistician	09/24/18 – 08/26/23
Extracorporeal Life Support Organization (PI Boonstra); \$40,000 <i>ELSO Registry Contract</i> Role: PI	10/01/19 – 12/31/21
Clinical Trial (PI Phillips); \$281,854 AbbVie, Inc. <i>Phase I multi-site study evaluating the MTD, safety and efficacy of the combination venetoclax, lenalidomide and rituximab in patients with previously untreated mantle cell lymphoma</i>	12/14/18 – 08/01/21

Role: Biostatistician	
1-R21HD090366, NIH/NICHD (PI West); \$275,000 <i>Indices of Selection Bias for Non-Probability Samples</i> Role: Co-I	09/23/17 – 08/31/20
MICHR Pathway Award (PI Zelner); \$50,000 <i>Integrated Studies of Vaccine Effectiveness and Failure</i> CTSA support from grant 5-UL1TR002240, NIH/NCATS Role: Co-I	09/01/19 – 08/31/20
2-P01CA059827, NIH/NCI (PI TenHaken); \$1,784,750 <i>Optimization of High Dose Conformal Therapy</i> Role: Co-I	05/15/14 – 04/30/19
MICHR Pilot Grant Program (PI Boonstra); \$15,000 <i>Measuring consensus in the problem list using penalized regression models for rank data</i> CTSA support from grant 5-UL1TR002240, NIH/NCATS Role: PI	03/01/18 – 02/28/19
5-U01DE025633, NIH/NIDCR (PI DaSilva); \$282,634 <i>Investigation and Modulation of the Mu-Opioid Mechanism in Chronic TMD (in vivo)</i> Role: Biostatistician	08/01/16 – 07/31/21
5-R01NS094413, NIH/NINDS (PI DaSilva); \$361,417 <i>Investigation and Modulation of the Central Mu-Opioid</i> Role: Biostatistician	09/30/15 – 08/31/20
Research Agreement, CTI BioPharma Corp. (PI Talpaz); \$75,032 <i>A New Prognostic Model for Response in Myelofibrosis Patients Treated with JAK2 Inhibitors</i> Role: Co-I	12/11/15 – 12/10/17
R6503-16, Leukemia and Lymphoma Society (PI Wilcox); \$270,027 <i>Peripheral T-cell Lymphoma, not otherwise specified: The cell of origin as a predictive biomarker and therapeutic target</i> Role: Biostatistician	10/01/15 – 09/30/17
Award letter dated 10/22/15, MCubed2.0 Executive Committee (PIs Schipper / Boonstra / Matuszak); \$60,000 <i>Identifying treatment*biomarker interactions by combining a large dose-toxicity dataset without biomarkers and a similar but smaller dataset with biomarkers</i> Role: Co-PI	10/22/15 – 04/29/17
Award letter dated 08/21/15, Prostate Cancer Foundation (PI Palapattu); \$600,000 <i>Testing Targeted NK (TaNK) cell Therapy in Prostate Cancer</i> Role: Co-I	09/01/15 – 09/01/17
Award letter dated 05/02/14, Cancer Research Committee, UMCCC (PI Boonstra); \$25,000 <i>Increasing efficiency for estimating treatment-biomarker interactions with historical data</i> Role: PI	05/01/14 – 04/30/15

Seminars and Panels – External

1. *The statistician – numbers that matter* 14th EuroELSO Congress, Dublin, Ireland, May 2026. **Invited Seminar**
2. *Dose-finding in Oncology Clinical Trials* Dordt University, Department of Mathematics & Statistics Speaker Series, Sioux Center, IA (virtual), Nov 2025. **Invited Seminar**
3. *Working with Industry Using the ELSO Registry* 36th Annual ELSO Conference, Oxon Hill, MD, Sept 2025. **Invited Seminar**
4. *Some novel clinical trial designs for dose-finding in oncology* Hope College, Department of Mathematics & Statistics Colloquium Series, Holland, MI, Oct 2024. **Invited Seminar**

5. *Some Challenges in Statistical Modeling in ECLS* 35th Annual ELSO Conference, Detroit, MI, Sept 2024. **Invited Seminar**
6. *Why you should consider a graduate education in biostatistics* Calvin University, Department of Mathematics & Statistics Colloquium, Grand Rapids, MI, Sept 2023. **Invited Seminar**
7. *Targeted randomization dose optimization trials enable fractional dosing of scarce drugs* The 36th New England Statistics Symposium, Boston, MA (virtual presenter), June 2023. **Invited Presentation**
8. *Industry use of the ELSO Registry to Support Device Approval and Surveillance* 33rd Annual ELSO Conference, Boston, MA, Sept 2022. **Invited Presentation**
9. *A comparison of methods for incorporating information from historical prediction models* 43rd Annual Conference of the International Society for Clinical Biostatistics, Newcastle, UK, Aug 2022. **Contributed Talk**
10. *ECMO for COVID-19: Evolving Outcomes from the ELSO Registry* 32nd Annual ELSO Conference (Virtual), Sept 2021. **Invited Panel**
11. *The ELSO Registry on ECMO for COVID-19* 31st Annual ELSO Conference (Virtual), Sept 2020. **Invited Panel**
12. *Statistics is not an easy-bake oven* Hope College, Department of Mathematics & Statistics Colloquium Series, Holland, MI, Mar 2020. **Invited Seminar**
13. *Measures of the degree of departure from ignorable sample selection* Joint Statistical Meetings, Denver, CO, July 2019. **Invited Talk**
14. *A modular framework for seamless oncology trials* Columbia University, Dept. of Biostatistics, New York, NY, Mar 2019. **Invited Seminar**
15. *A default prior for the intercept parameter in logistic regression* Joint Statistical Meetings, Vancouver, BC, Aug 2018. **Topic Contributed Talk**
16. *Incorporating historical models with adaptive Bayesian updates* 2018 WNAR Meeting, Edmonton, AB, June 2018. **Contributed Talk**
17. *Combining nested risk-prediction models* 38th Annual Conference of the International Society for Clinical Biostatistics, Vigo, Spain, July 2017. **Contributed Talk**
18. *A default prior for the intercept in binary-data regression models* 2017 ENAR Spring Meeting, Washington, DC, Mar 2017. **Contributed Talk**
19. *Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts* 5th Early Phase Adaptive Trials Workshop, Politecnico di Torino, Turin, Italy, Oct 2016. **Invited Talk**
20. *Efficient designs for dose-expansion in phase I oncology trials* Fred Hutchinson Cancer Research Center, Seattle, WA, Oct 2015. **Invited Seminar**
21. *Leveraging historical data for improved estimates of treatment-biomarker interactions* 36th Annual Conference of the International Society for Clinical Biostatistics, Utrecht, The Netherlands, Aug 2015. **Contributed Talk**
22. *Strategies for tuning parameter selection in regression problems with many covariates* Fred Hutchinson Cancer Research Center, Seattle, WA, Jan 2015. **Invited Seminar**
23. *A statistical evaluation of expansion cohorts in phase I clinical trials* 4th Early Phase Adaptive Trials Workshop, Medical University of South Carolina, Charleston, SC, Oct 2014. **Invited Talk**
24. *Increasing efficiency for estimating treatment-biomarker interactions with historical data* 2014 Midwest Biopharmaceutical Statistics Workshop, Ball State University, Muncie, IN, May 2014. **Contributed Talk**
25. *Prediction with high-dimensional data: A brief foray into biostatistics* Hope College, Department of Mathematics Colloquium Series, Holland, MI, Oct 2013. **Invited Seminar**

26. *Adaptive shrinkage via the hyperpenalized EM algorithm* 34th Annual Conference of the International Society for Clinical Biostatistics, Munich, Germany, Aug 2013. **Contributed Talk**
27. *Adaptive shrinkage via the hyperpenalized EM algorithm* Joint Statistical Meetings, Montreal, QC, Aug 2013. **Invited Talk**
28. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Joint Statistical Meetings, San Diego, CA, Aug 2012. **Contributed Talk**
29. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Highlighted Student Talk, Michigan Student Symposium in Statistical Science, Ann Arbor, MI, Apr 2012. **Contributed Talk**
30. *Prediction with high-dimensional data: A brief foray into biostatistics* Calvin College, Department of Mathematics & Statistics Colloquium, Grand Rapids, MI, Dec 2011. **Invited Seminar**
31. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Conference on Risk Assessment and Evaluation of Predictions, Silver Spring, MD, Oct 2011. **Contributed Talk**
32. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Joint Statistical Meetings, Miami Beach, FL, Aug 2011. **Contributed Talk**
33. *Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in Lynch Syndrome* Joint Statistical Meetings, Vancouver, BC, Aug 2010. **Contributed Talk**

Seminars and Panels – Internal

1. *Some novel extensions of the horseshoe prior* University of Michigan, Dept. of Biostatistics, Ann Arbor, MI, Jan 2020. **Invited Seminar**
2. *Incorporating historical information with adaptive Bayesian updates* Workshop on Bayes, Fiducial, and Frequentist Paradigm in Data Integration, Machine Learning, and Applications, University of Michigan, Dept. of Biostatistics, Ann Arbor, MI, Nov 2019. **Invited Seminar**
3. *A modular framework for seamless oncology trials* University of Michigan, Dept. of Radiation Oncology Biostatistics Division Seminar, Ann Arbor, MI, Dec 2018. **Invited Seminar**
4. *Modeling consensus among ordered lists using variable selection penalties* Rod Little Lectureship Dinner, Ann Arbor, MI, Sep 2018. **Invited Speed Talk**
5. *Fifth Bayesian, Fiducial, and Frequentist (BFF5) Conference* University of Michigan, Ann Arbor, MI, May 2018. **Invited Panelist**
6. *Combining Bayesian generalized linear models with map-weighting* Biomedical Statistical Modeling Conference, Ann Arbor, MI, June 2017. **Invited Talk**
7. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Department of Biostatistics Seminar, University of Michigan, Ann Arbor, MI, Nov 2012. **Invited Seminar**
8. *Incorporating auxiliary information for improved prediction in high dimensional datasets: An ensemble of shrinkage approaches* Department of Statistics Student Seminar, University of Michigan, Ann Arbor, MI, Oct 2012. **Invited Seminar**
9. *A review of statistical methods for testing genetic anticipation: looking for an answer in Lynch Syndrome* Molecular Epidemiology of Colorectal Cancer Investigators' Meeting, Ann Arbor, MI, Oct 2009 and June 2010.

Workshops and Guest Lectures

1. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2025. **Guest Lecture / Tutorial**
2. *Clinical Trial Statistics for Clinical Investigators* Hematology-Oncology Fellows, Rogel Cancer Center, July 2025. **Guest Lecture**
3. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2024. **Guest Lecture**
4. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2024. **Guest Lecture**
5. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2023. **Guest Lecture**
6. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2023. **Guest Lecture**
7. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2022. **Guest Lecture**
8. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2022. **Guest Lecture**
9. *Teaching R with rstudio.cloud* University of Michigan, Dept. of Biostatistics, BIOS 803: Pedagogical Methods For Biostatistics Courses, Ann Arbor, MI, Feb 2022. **Guest Lecture**
10. *De-escalation designs for finding the minimum effective dose of a drug* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Sept 2021. **Guest Lecture**
11. *Introduction to R* University of Michigan, MICHR Bayesian Workshop Series, Ann Arbor, MI, Aug 2019. **Workshop**
GitHub repo
12. *Introduction to R markdown* University of Michigan Big Data Summer Institute, Ann Arbor, MI, July 2019. **Guest Lecture**
GitHub repo
13. *Assessment of predictive models* University of Michigan Big Data Summer Institute, Ann Arbor, MI, June 2019. **Guest Lecture**
14. *Expanding your biostatistics toolkit for biomedical research* University of Michigan, Global StatCore Workshop, Ann Arbor, MI, May 2019. **Workshop**
GitHub repo
15. *A modular framework for seamless oncology trials* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Nov 2018. **Guest Lecture**
16. *Introductory clinical trials* Hematology-Oncology Fellows, Rogel Cancer Center, July 2018. **Guest Lecture**
17. *Expanding your biostatistics toolkit for biomedical research* University of Michigan, Global StatCore Workshop, Ann Arbor, MI, May 2018. **Workshop**
18. *Building your biostatistics toolbox for cancer research* Dharmas Cancer Hospital, West Jakarta, Indonesia, Aug 2017. **Workshop**
19. *Building the bridge to phase II: Efficacy estimation in dose-expansion cohorts* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Sep 2016. **Guest Lecture**

20. *Building your biostatistics toolbox: Clinical Trials* St. Pauls Hospital Medical Millennium College, Addis Ababa, Ethiopia, Mar 2016. **Workshop**
21. *Building your biostatistics toolbox: Clinical Trials* Kenya Medical Research Institute, Nairobi, Kenya, Mar 2016. **Workshop**
22. *Adaptive shrinkage via the hyperpenalized EM algorithm* University of Michigan, Dept. of Biostatistics, BIOS 803: Biostatistics in Cancer Seminar, Ann Arbor, MI, Oct 2013. **Guest Lecture**
23. *Bayesian modeling for genetic anticipation in presence of mutational heterogeneity: A case-study in Lynch Syndrome* University of Michigan, Dept. of Biostatistics, BIOS 682: Applied Bayesian Inference, Ann Arbor, MI, Apr 2011. **Guest Lecture**

Service Activities

Internal

Member, Student Recruitment Committee Department of Biostatistics, UM	Fall 2024 – ongoing
Chair, Curriculum Committee Department of Biostatistics, UM	Fall 2021 – Summer 2023; Fall 2024 – ongoing
Member, Protocol Review Committee Rogel Cancer Center, UM	Winter 2016 – ongoing
Reviewer, MICHR Clinical and Translational Science Pilot Award Michigan Institute for Clinical and Health Research, UM	Fall 2025
Member, Advisory Committee on Academic Programs School of Public Health, UM	Fall 2021 – Summer 2023; Fall 2024 – Summer 2025
Member, Qualifying Exam Committee Department of Biostatistics, UM	2015–2016, 2020–2021, 2022–2023
Member, Cancer Data Science Faculty Search Committee Department of Biostatistics, UM	Fall 2021
Reviewer, MICHR Pilot Grant Program – Promoting Progress in Statistics Award Michigan Institute for Clinical and Health Research, UM	Fall 2021, Fall 2018
Statistician, Hem-Onc Fellows' journal club Rogel Cancer Center, UM	Feb 2018, Jan 2017, Aug 2015, Feb 2015, Apr 2014, Aug 2013, Mar 2013
Organizing Committee, Biomedical Statistical Modeling Conference Department of Biostatistics, UM	June 2017
Chair, Seminar Committee Department of Biostatistics, UM	Fall 2014 – Winter 2015
Faculty Advisor, Student Brownbag Committee Department of Biostatistics, UM	Fall 2014 – Winter 2015
Member, Bioinformatics/Statistical Genetics Faculty Search Committee Department of Biostatistics, UM	Fall 2010 – Winter 2011

External

Ad-hoc peer review service

American Journal of Clinical Nutrition (2015)
Annals of Applied Statistics (2020)
Bayesian Analysis (2017,2015,2014)

Bioinformatics (2016)
Biometrics (2026,2021,2019,2018,2017,2016,2015,2012)
Blood (2023)
BMC Bioinformatics (2016)
BMC Research Notes (2020,2017)
BMJ Open (2018,2017)
British Journal of Radiology (2017)
Cancer (2025,2020)
Cancer Cytopathology (2018,2017,2016)
Circulation: Cardiovascular Quality and Outcomes (2024,2019,2018)
Clinical Trials (2019,2017,2016)
Computational Statistics & Data Analysis (2012)
Communications in Statistics (2024)
Contemporary Clinical Trials Communications (2018)
Electronic Journal of Statistics (2014)
eLife (2026,2025,2024,2023,2022,2021)
Genetic Epidemiology (2011)
Harvard Data Science Review (2021)
Heliyon (2023)
IEEE Transactions on Cybernetics (2022,2014)
Iranian Journal of Science and Technology, Transactions A: Science (2020,2019)
International Journal of Medical Informatics (2020)
International Journal of Radiation Oncology · Biology · Physics (2020,2019,2018,2017,2016,2015,2014)
International Statistical Review (2024,2023,2020,2013)
JAMA Dermatology (2017)
JAMA Oncology (2025,2023,2022,2012)
JCI Insight (2020)
Journal of Agricultural, Biological, and Environmental Statistics (2015,2014,2013)
Journal of the American Medical Association (JAMA) (2014)
Journal of the American Statistical Association (JASA) (2019,2018)
Journal of Multivariate Analysis (2018)
Journal of Official Statistics (2025)
Journal of the National Cancer Institute (2020)
Journal of the Royal Statistical Society: Series A (JRSSA) (2024,2023)
Journal of the Royal Statistical Society: Series C (JRSSC) (2026,2025)
Journal of Statistical Computation and Simulation (2025,2024)
Journal of Statistical Planning & Inference (2017,2013)
Journal of Survey Statistics and Methodology (2023)
Nature Medicine (2023)
Pharmaceutical Statistics (2019)
PLOS One (2024,2019,2018,2014)
Psychological Methods (2023)
Scientific Reports (2021,2020)
Scientific World Journal (2014)
Statistica Sinica (2013)
Statistical Methods in Medical Research (2022,2021,2020,2019)
Statistics in Medicine (2021,2020,2018,2017,2014,2013)

The American Statistician (2018,2017)
The Annals of Applied Statistics (2020)
The Lancet (2024,2022,2018)
The Lancet Digital Health (2025)
The Lancet Haematology (2026,2025,2023,2021,2020)
The Lancet Psychiatry (2024,2023,2022,2021,2020,2019)

Other service activities

AP Statistics Presenter, Saline High School May, 2022–2026, Saline, MI
 Reviewer, National Institutes of Health Apr 2026, Virtual
 Clinical Oncology (CONC) Study Section, National Cancer Institute
 Member, Data Safety Monitoring Board 2024 – ongoing
 ED-LEAD: Emergency Departments Leading the Transformation of Alzheimer’s and Dementia Care
 (clinicaltrials.gov NCT06079203)
 Reviewing Editor, eLife journal 2021 – ongoing
 Reviewer, ASA Health Policy Statistics Section Dec 2025
 2026 HPSS Student Paper Competition
 AP Statistics Presenter, Pioneer High School May 2025, Ann Arbor, MI
 Executive Committee Member, International Society for Clinical Biostatistics (ISCB) Jan 2023 – Dec 2024
 Session Chair 2024 International Society for Clinical Biostatistics, Thessaloniki, Greece
 Biostatistical Reviewer, American Society of Clinical Oncology Feb 2024, Arlington, VA
 2024 Grants Selection Committee – Young Investigator Awards, Panel B
 Reviewer, ASA Biometrics Section Winter 2024
 2024 David P. Byar Young Investigator Award
 Oral Session Chair, 44th Annual Conference of the International Society for Clinical Biostatistics,
 Milan, Italy, Aug 2023
 External Reviewer, Graduate School of Social Sciences, Vrije Universiteit Amsterdam Dec 2022
 Reviewer for go/no-go assessment of Santiago Gómez-Echeverry’s PhD project
 Specialist Reviewer, Department of Defense Nov 2022, Virtual
 2022 Prostate Cancer Research Program, DoD Congressionally Directed Medical Research Programs
 Member, Data Safety Monitoring Board 2017– 2022
 mHealth to Improve Blood Pressure Control in Hypertensive African Americans (MI-BP)
 (clinicaltrials.gov NCT02955537)
 Specialist Reviewer, Department of Defense Sep 2021, Virtual
 2021 Prostate Cancer Research Program, DoD Congressionally Directed Medical Research Programs
 Member, Data Safety Monitoring Board 2019– 2020
 Fibromyalgia and Morning Light Treatment study (Fibrolight)
 (clinicaltrials.gov NCT03794908)
 Reviewer, National Institutes of Health Feb 2020, San Diego, CA
 Cancer Biomarkers Study Section, National Cancer Institute
 Reviewer, AVIESAN and the French National Cancer Institute Aug 2019
 Interdisciplinary approaches in oncogenic processes and therapeutic perspectives: Contributions of mathematics and informatics to oncology
 Reviewer, National Institutes of Health Apr 2019, Bethesda, MD
 HEAL Initiative: Justice Community Opioid Innovation Network (JCOIN) Clinical Research Centers
 Reviewer, ASA Biometrics Section Winter 2019
 2019 David P. Byar Young Investigator Award
 Invited Academic Guest Editor, PLOS One Winter 2019, Fall 2018

Discussant, National Institutes of Health	Mar 2018, Washington, DC
Senator Paul D. Wellstone Muscular Dystrophy Specialized Research Centers (MDSRCs)	
JSM Contributed Session Chair	2014 Joint Statistical Meetings, Boston, MA
JSM Docent	2014 Joint Statistical Meetings, Boston, MA
JSM Invited Session Chair	2012 Joint Statistical Meetings, San Diego, CA
JSM Contributed Session Chair	2011 Joint Statistical Meetings, Miami Beach, FL

Memberships

American Statistical Association	since 2010
International Biometrics Society (Eastern North American Region)	since 2011
International Society for Clinical Biostatistics	since 2013